Brody M

HW 5

Stat 2112

14.24). a.) $t=2$ $L_2=3.5$ $b_2=.25$

$t=3 \rightarrow x_3=3.9$

$L_3=.2(3.9)+.8(3.5+.25)=3.78$ $\quad Q_3 \approx .27$

$b_3=.6(3.78-3.5)+.4(.25)=.268$

b.) $t=4$ $x_4=4.25$ $L_3=3.78$ $b_3=.268$

$L_4=.2(4.25)+.8(3.78+.268)=4.088$

$b_4=.6(4.088-3.78)+.4(.268)=.2922$ $\quad Q_4 \approx 4.09$

c.) $x_5=L_4+1\cdot b_4=4.3806$

$Q_5 \approx 4.38$

14.31) a.) $w=.5$

$S_{Jan\,2015}=x_{Jan\,2015}=1250.75$

Dec 2018: $SES_Dec\,2018=1201.24$

$SES\ Dec\,2018=1201.24$

b.) 2019:

Month	SES-one step forecast	Actual
Jan	1201.24	1291.75
Feb	1256.75	1320.07
Mar	1288.41	1300.90
Apr	1294.65	1285.91
May	1290.28	1283.7
Jun	1286.99	1359.04
Jul	1322.01	1412.89
Aug	1367.45	1500.41
Sep	1433.43	1510.58

Brody M. HW5

14.31) continued)

c.) $\alpha = .5 \quad \beta = .5$

$L_{jan2015} = x_{jan2015} = 1250.75$

$1227.08 - 1250.75 = -23.67$

Dec 2018:

$L_{Dec_{2018}} = 1220.32$

$b_{Dec\,2018} = 9.88$

one step forcast $\overset{2018}{Dec} - \overset{2019}{Jan} = 1230.25$

14.32 a.) $\alpha = .3$

$F_{2004} = 5.83$

$F_{2005} = .3 \cdot X_{2004} + .7 \cdot F_{2004} = .3 \cdot 5.83 + .7 \cdot 5.83 = 5.83$

$F_{2006} = .3 \cdot X_{2005} + .7 \cdot F_{2005} = 6.016$

$F_{2007} = .3 \cdot (7.10) + .7(6.016) = 6.427$

c.)
MAD = 3.970M Barrells
MAPE (23.58)
RMSE = 4.85M Barrells

b.) $\alpha = .7 \quad F_{2004} = 5.83 \quad F_{2005} = 5.83$

$F_{2006} = 6.05 \quad F_{2007} = 6.83$

d.
MAD = 1.92M Barrells
MAPE = 12.54%
RMSE = 2.646M Barrells

Brody M

Stat 1112 HW5

14.32 continued)

(e.) All three accuracy measures are smaller for $\alpha = .7$ meaning exponentially smoothed forecasts track better than $\alpha .3$ (one-step forecasts)

f.) you should use $\alpha = .7$